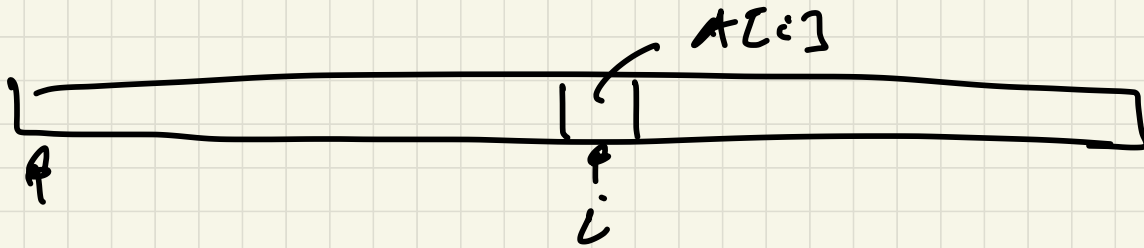



MINIMUM IN AN ARRAY

Input: A array $|A| = n$

Output: the minimum value



Complete linear scan

INPUT SIZE $n = |A|$

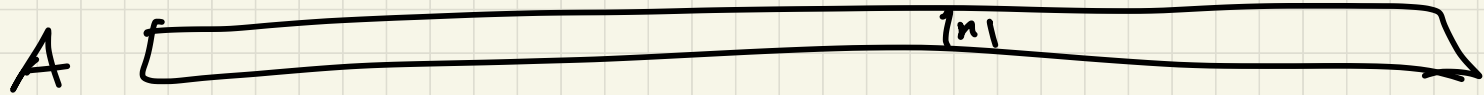
TIME COMPLEXITY LINEAR IN n

~
ALWAYS

SEARCH IN AN ARRAY

Input: A array $|A| = n$, k value

Output: i . $A[i] = k$, -1 on



- - - linear scan

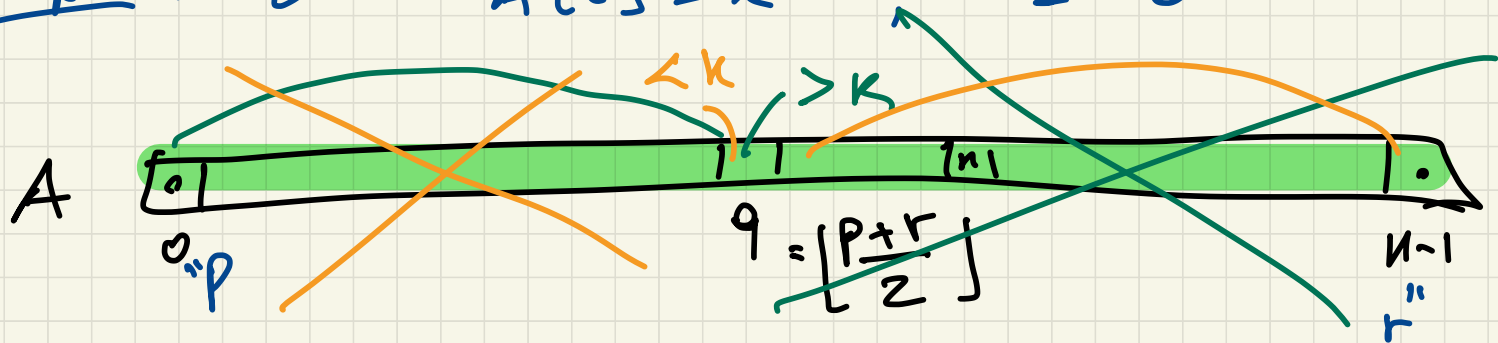
TIME COMPLEXITY

LINEAR

}
IN WORST CASE

SEARCH IN A SORTED ARRAY

Input: A ~~sorted~~ array $|A| = n$, k value
Output: i . $A[i] = k$ -1 onw



BINARYSEARCH($A, P, r, n-1, k$)

$\log_2 n$

1
2
4
⋮
n

$2^{\log_2 n} = n$

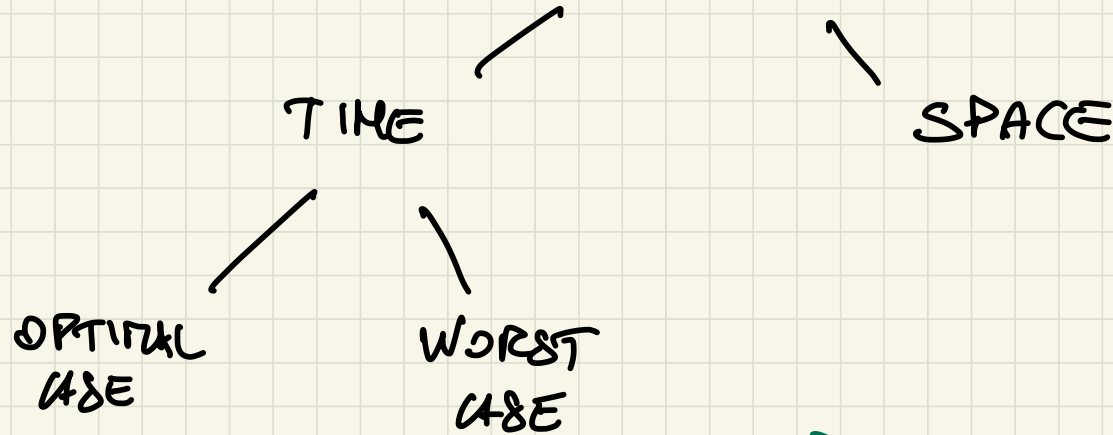
→

$A[q] = k$ ✓

$A[q] > k$ ●

$A[q] < k$ ●

COMPUTATION COST OF AN ALGORITHM



$$T(n) = \cancel{8n^2} + 5n + \cancel{\log n} + 4$$

the function that describes your time complexity

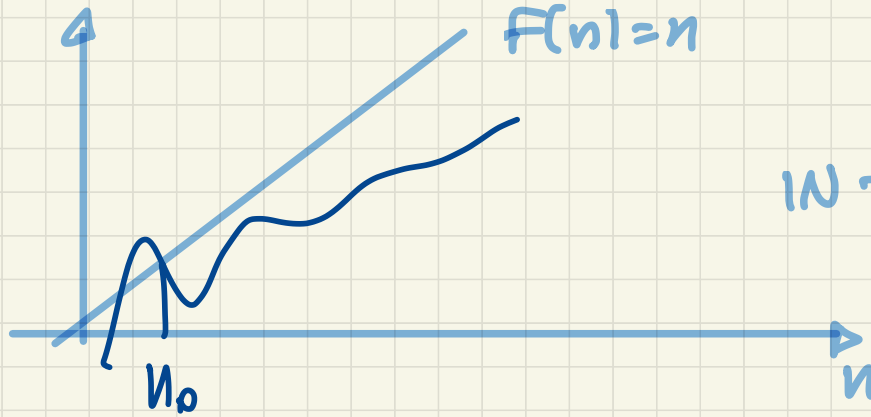
CONSTANT
MULTIPLICATIVE
FACTOR

SMALLER TERMS

A QUADRATIC
TIME COMPLEXITY

ASYMPTOTIC NOTATION

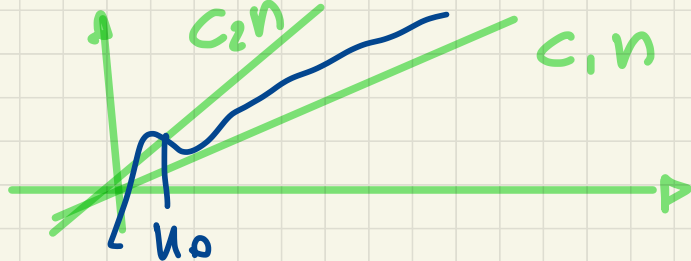
TIME COMPLEXITY IN



$$\mathcal{O}(n)$$

IN THE WORST CASE

TIME COMPLEXITY IN



$$\Theta(n)$$

"ALWAYS, LINEAR"

PROBLEM SOLVING

Input: Array of integers

Output: TRUE if $\exists i \quad A[i] = i$
FALSE otherwise



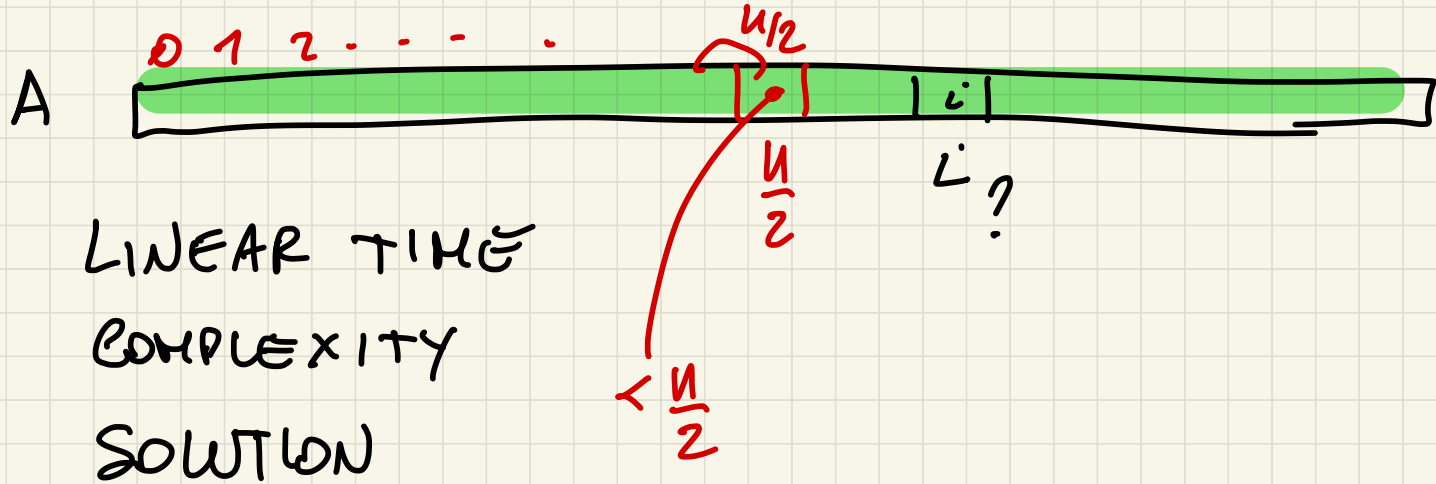
LINEAR TIME

COMPLEXITY

SOLUTION

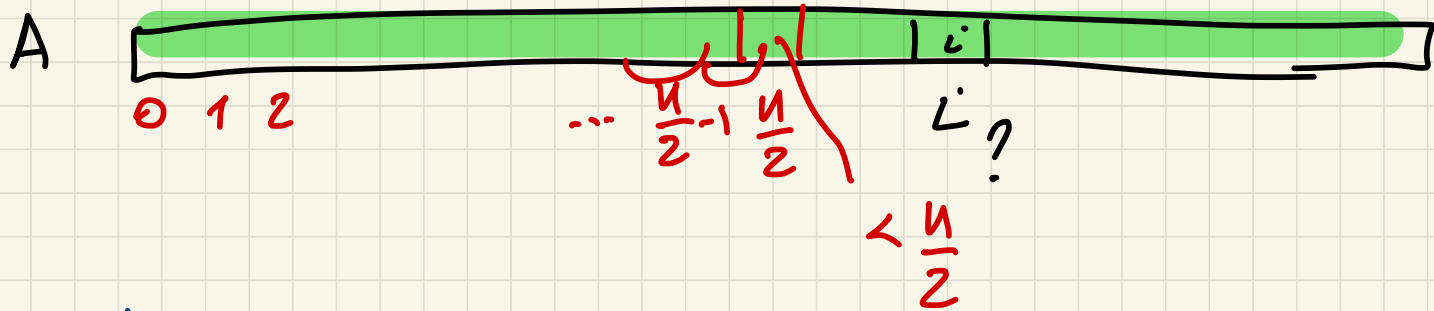
Input : SORTED array A of integers

Output : TRUE if $\exists i \quad A[i] = i$
FALSE otherwise



Input: SORTED array A of distinct integers

Output: TRUE if $\exists i \text{ } A[i] = i$
FALSE otherwise

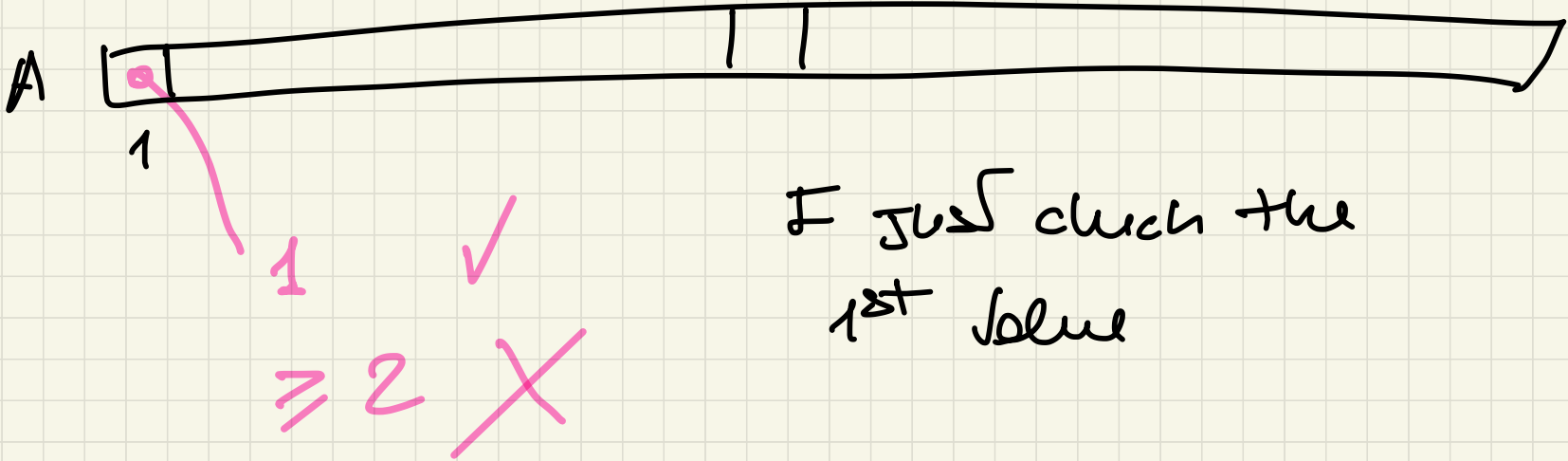


A SORT OF
BINARY SEARCH
SOLUTION

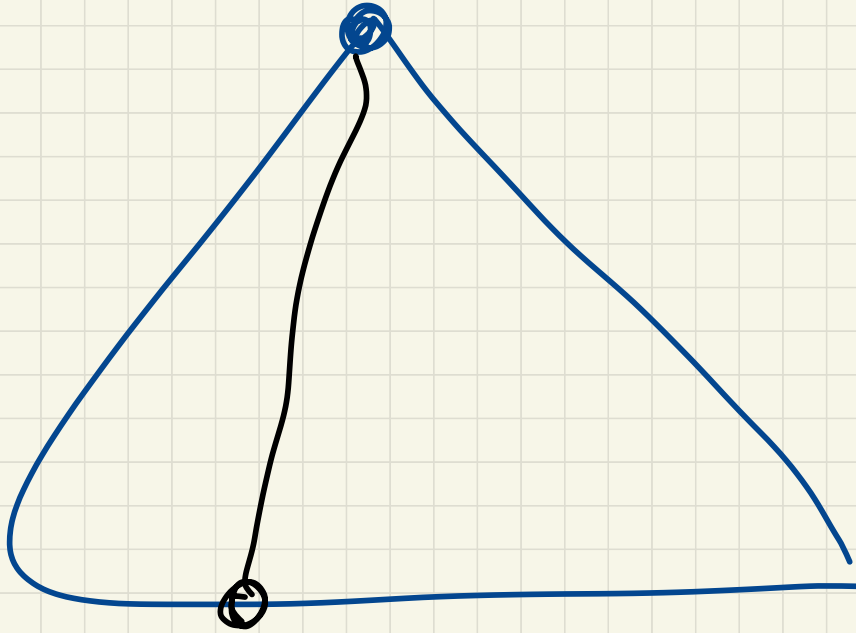
$O(\log n)$
COMPLEXITY

Input: SORTED array A of distinct positive

Output: TRUE if $\exists i \text{ } A[i] = i$ integers
FALSE otherwise



TREES



n nodes



height of
a balanced
tree $\log n$

GRAPHS

n # nodes
 m # edges
($m \in O(n^2)$)

$m \in \Theta(n)$ SPARSE GRAPH
 $m \in \Theta(n^2)$ DENSE GRAPH

'degree' of a node

$|\Sigma|$ size of alphabet

